Binomial Theorem（二項式定理）

Before introducing Binomial Theorem, let's have a look on **Pascal's Triangle (帕斯卡三角形)** and **Binomial Coefficient (二項式係數)**:

帕斯卡三角形
Pascal's Triangle

$$1$$
$$1 \quad 1$$
$$1 \quad 2 \quad 1$$
$$1 \quad 3 \quad 3 \quad 1$$
$$1 \quad 4 \quad 6 \quad 4 \quad 1$$

二項式係數
Binomial Coefficient

$$1$$
$$C_0^1 \quad C_1^1$$
$$C_0^2 \quad C_1^2 \quad C_2^2$$
$$C_0^3 \quad C_1^3 \quad C_2^3 \quad C_3^3$$
$$C_0^4 \quad C_1^4 \quad C_2^4 \quad C_3^4 \quad C_4^4$$

The definition of Binomial Coefficient us $C_r^n = \dfrac{n!}{r!(n-r)!}$, where $n! = n(n-1)(n-2)...3 \cdot 2 \cdot 1$

But there is an easier way to compute binomial coefficients:

$$C_1^5 = \frac{5}{1} = 5, \quad C_2^5 = \frac{5 \cdot 4}{1 \cdot 2} = 10, \quad C_3^5 = \frac{5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3} = 10$$

$$C_1^6 = \frac{6}{1} = 6, \quad C_2^6 = \frac{6 \cdot 5}{1 \cdot 2} = 15, \quad C_3^6 = \frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} = 20$$

In general,

$$C_1^n = \frac{n}{1} = n \qquad C_2^n = \frac{n(n-1)}{2} \qquad C_3^n = \frac{n(n-1)(n-2)}{6}$$

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Binomial Theorem（二項式定理）

The binomial $(1 + x)^n$ can be expanded with the use of C_r^n as follows:

$$(1+x)^0 = 1$$

$$(1+x)^1 = 1+x$$

$$(1+x)^2 = 1+2x+x^2$$

$$(1+x)^3 = 1+3x+3x^2+x^3$$

$$(1+x)^4 = 1+4x+6x^2+4x^3+x^4$$



$$(1+x)^5 =$$

$$(1+x)^6 =$$

The binomial $(a + b)^n$ can be expanded with the use of C_r^n as follows:

$$(a+b)^0 = 1$$

$$(a+b)^1 = a+b$$

$$(a+b)^2 = a^2+2ab+b^2$$

$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$

$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$



$$(a+b)^5 =$$

$$(a+b)^6 =$$

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Binomial Theorem



[Self-designed question]

(a) Evaluate the value of C_1^9, C_2^9, C_3^9 and C_4^9 respectively.

(b) Expand $(1 + x)^9$ in ascending powers of x up to the term x^4.
Hence estimate the value of $(1.2)^9$ without using calculator.

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Binomial Theorem

The following is **Binomial Theorem (二項式定理)** :

$$(a+b)^n = C_0^n a^n b^0 + C_1^n a^{n-1} b^1 + C_2^n a^{n-2} b^2 + \ldots + C_n^n a^{n-n} b^n$$

OR $$(a+b)^n = a^n + C_1^n a^{n-1} b^1 + C_2^n a^{n-2} b^2 + \ldots + b^n$$

OR $$(a+b)^n = \sum_{r=0}^{n} C_r^n a^{n-r} b^r$$

e.g.1 Expand $(x + 2y)^3$.

$$\begin{aligned}(x+2y)^3 &= x^3 + 3(x^2)(2y) + 3(x)(2y)^2 + (2y)^3 \\ &= x^3 + 6x^2y + 12xy^2 + 8y^3\end{aligned}$$

多加括號可防二項式展開時有錯

e.g.2 Expand $(x – 2y)^3$.

$$\begin{aligned}(x-2y)^3 &= x^3 + 3(x^2)(-2y) + 3(x)(-2y)^2 + (-2y)^3 \\ &= x^3 - 6x^2y + 12xy^2 - 8y^3\end{aligned}$$

二項式中間是負號，展開後答案是會出現「正負正負…」規律

e.g.3 Expand $(x^2 - \frac{1}{x})^4$

$$\begin{aligned}(x^2-\frac{1}{x})^4 &= (x^2)^4 - 4(x^2)^3(\frac{1}{x}) + 6(x^2)^2(\frac{1}{x})^2 - 4(x^2)(\frac{1}{x})^3 + (\frac{1}{x})^4 \\ &= x^8 - 4x^5 + 6x^2 - \frac{4}{x} + \frac{1}{x^4}\end{aligned}$$

除了二項式係數外，代數上的指數亦會有一定規律 (等差數列)

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Binomial Theorem

[A.Math 1995 Paper II Q.4]

Given $(x^2+\frac{1}{x})^5-(x^2-\frac{1}{x})^5=ax^7+bx+\frac{c}{x^5}$,

find the values of *a*, *b* and *c*. Hence, evaluate $(2+\frac{1}{\sqrt{2}})^5-(2-\frac{1}{\sqrt{2}})^5$.

(6 marks)

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Binomial Theorem

二項式兩大特別題型（一）： *Binomial × Quadratic Equation*

In the expansion of $(a+b)^n$,

$$(a+b)^n = a^n + C_1^n\, a^{n-1} b^1 + C_2^n\, a^{n-2} b^2 + \ldots$$

where $C_1^n = n$ and $C_2^n = \dfrac{n(n-1)}{2}$

e.g. In the expansion of $(1-3x)^n + (1+x)^n$, where n is a positive integer, the coefficient of x^2 is 150. Find the value(s) of n.

$$(1-3x)^n = 1 + C_1^n(-3x) + C_2^n(-3x)^2 + \ldots$$
$$= 1 - 3nx + \frac{9n(n-1)}{2}x^2 + \ldots$$

緊記 $C_2^n = \dfrac{n(n-1)}{2}$

$$(1+x)^n = 1 + C_1^n x + C_2^n x^2 + \ldots$$
$$= 1 + nx + \frac{n(n-1)}{2}x^2 + \ldots$$

For the x^2 term,

$$\frac{9n(n-1)}{2} + \frac{n(n-1)}{2} = 150$$
$$5n(n-1) = 150$$
$$n^2 - n - 30 = 0$$

將 C_2^n 變成二次方

$$(n-6)(n+5) = 0$$
$$n = 6 \quad \text{or} \quad -5 \text{ (rejected)}$$

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Binomial Theorem

[A.Math 1986 Paper II Q.2]

In the expansion of $(x^2+2)^n$ in descending powers of x, where n is a positive integer, the coefficient of the third term is 40. Find the value of n and the coefficient of x^4. ***(5 marks)***

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Binomial Theorem

二項式兩大特別題型（二）：「二項式通項 General Term」

In the expansion of $(a+b)^n$,

$$\text{General term} = C_r^n \, a^{n-r} \, b^r$$

e.g. In the expression $(2x^2 - \frac{1}{x})^6$, find

(i) the term in x^3,

(ii) the constant term (常數項).

In the expression $(2x^2 - \frac{1}{x})^6$,

$(r+1)^{\text{th}}$ term $= C_r^6 (2x^2)^{6-r} (-\frac{1}{x})^r$ ← 使用二項式通項

$= C_r^6 (2)^{6-r} \cdot (x^2)^{6-r} \cdot (-1)^r \cdot (\frac{1}{x})^r$

$= C_r^6 (-1)^r \cdot (2)^{6-r} \cdot x^{12-2r} \cdot x^{-r}$

$= C_r^6 (-1)^r \cdot (2)^{6-r} \cdot x^{12-3r}$ ← 留意 x 的係數 (這一次是 **$12 - 3r$**)

(i) Set **$12 - 3r = 3$**, ∴ **$r = 3$** ← 設 x 的係數是 3, ∴ **$12 - 3r = 3$**

∴ the term in x^3 $= C_3^6 (-1)^3 \cdot (2)^{6-3} \cdot x^3$

$= -160 \, x^3$

(ii) Set **$12 - 3r = 0$**, ∴ **$r = 4$** ← 常數代表 x 的係數是 0, ∴ **$12 - 3r = 0$**

∴ constant term $= C_4^6 (-1)^4 \cdot (2)^{6-4} \cdot x^0$

$= 60$

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Binomial Theorem

[A.Math 2003 Q.12]

Determine whether the expansion of $(2x^2 + \frac{1}{x})^9$ consists of

(a) a constant term,

(b) an x^2 term.

In each part, find the term if it exists. ***(6 marks)***

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Binomial Theorem (Double Binomial)



二項式兩大特別題型（三）：

" OVER THE RAINBOW"

– Double Binomial 雙重二項式

雙重二項式中，不要硬爆括號！

e.g. Find the coefficient of x^2 in the expansion of $(1+2x)^3(1-x)^6$

$(1+2x)^3(1-x)^6$

$= (1+C_1^3\cdot 2x+C_2^3\cdot(2x)^2+\ldots)(1+C_1^6(-x)+C_2^6(-x)^2+\ldots)$

$= (1+6x+12x^2+\ldots)(1-6x+15x^2+\ldots)$

Over the Rainbow：

選出所有相乘後是 x^2 的配搭出來！

the term in x^2

$= 1\cdot(15x^2)+6x\cdot(-6x)+12x^2\cdot(1)$

緊記不要漏正負號

$= -9x^2$

$\therefore$ the coefficient of $x^2 = -9$

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Binomial Theorem (Double Binomial)

[A.Math 2000 Paper II Q.2]

Expand $(1 + 2x)^7 (2 – x)^2$ in ascending powers of x up to the term x^2.

(5 marks)

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HKDSE Questions (level 1)

1. ***[M2 2013 Q.2]***

Suppose the coefficients of x and x^2 in the expansion of $(1+ax)^n$ are –20 and 180 respectively. Find the values of a and n.

(4 marks)

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HKCEE Questions (Level 1)

2. ***[A.Math 1985 Paper II Q.1]***

$(ax+\frac{1}{x^2})^n$ is expanded in descending powers of x, where n is a positive integer and $a > 0$. If the fourth term of expansion is constant which is equal to $\frac{21}{2}$, find the value of n and a.

(5 marks)

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HKCEE Questions (Level 1)

3. ***[A.Math 2002 Q.1]***

If n is a positive integer and the coefficient of x^2 in the expansion of $(1+x)^n + (1+2x)^n$ is 75, find the value(s) of n.

(8 marks)

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HKCEE Questions (Level 1)

4. ***[A.Math 1998 Paper II Q.1]***

Find the coefficient of x^2 in the expansion of $(x-\frac{2}{x})^6$. ***(4 marks)***

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HKCEE Questions (level 1)

5. ***[A.Math 2001 Q.4]***

Find the constant term in the expansion of $(2x^3 + \frac{1}{x})^8$. ***(4 marks)***

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HKDSE Questions (level 2)

6. ***[M2 2016 Q.1]***

Expand $(5+x)^4$. Hence, find the constant term in the expansion of

$(5+x)^4(1-\frac{2}{x})^3$. ***(5 marks)***

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HKDSE Questions (level 2)

7. ***[M2 2018 Q.2]***

Expand $(x+3)^5$. Hence, find the coefficient of x^3 term in the expansion of

$(x+3)^5(x-\frac{4}{x})^2$. ***(5 marks)***

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HKDSE Questions (level 2)

8. ***[M2 2020 Q.1]***

(a) Expand $(1-x)^4$.

(b) Find the constant k such that the coefficient of x^2 term in the expansion of $(1+kx)^9(1-x)^4$ is –3. ***(4 marks)***

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HKCEE Questions (level 3)

9. ***[A.Math 2004 Q.2]***

(a) Expand $(1+2x)^6$ in ascending powers of x up to the term x^3.

(b) Find the constant term in the expansion of $(1-\frac{1}{x}+\frac{1}{x^2})(1+2x)^6$.

(4 marks)

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HKCEE Questions (Level 3)

10. ***[A.Math 1999 Paper II Q.7]***

(a) Expand $(1 + 2x)^n$ in ascending powers of x up to the term x^3, where n is a positive integer.

(b) In the expansion of $(x - \frac{3}{x})^2(1 + 2x)^n$, the constant term is 210. Find the value of n.

(6 marks)

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HKCEE Questions (Level 3)

11. ***[A.Math 1992 Paper II Q.2]***

In the expansion of $(1 + 3x)^2 (1 + x)^n$, where n is a positive integer, coefficient of x is 10.

(a) Find the value of n.

(b) Find the coefficient of x^2. ***(5 marks)***

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HKDSE Questions (level 3)

12. ***[M2 2014 Q.1]***

In the expansion of $(1-4x)^2(1+x)^n$, the coefficient of x is 1.

(a) Find the value of n.

(b) Find the coefficient of x^2.

(4 marks)

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HKDSE Questions (level 3)

13. ***[M2 2021 Q.3]***

The coefficient of x^2 in the expansion of $(1-4x)^n$ is 240, where *n* is a positive integer. Find

(a) *n*,

(b) the coefficient of x^4 in the expansion of $(1-4x)^n\left(1+\frac{2}{x}\right)^5$.

(6 marks)

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HKCEE Questions (Level 4)

14. ***[A.Math 2005 Q.2]***

(a) Expand $(1+y)^5$.

(b) Using (a), or otherwise, expand $(1+x+2x^2)^5$ in ascending powers of x up to the term x^2.

(4 marks)

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HKDSE Questions (level 4)

15. ***[M2 2017 Q.2]***

Let $(1+ax)^8 = \sum_{k=0}^{8} \lambda_k x^k$ and $(b+x)^9 = \sum_{k=0}^{9} \mu_k x^k$, where a and b are constants. It is given that $\lambda_2 : \mu_7 = 7:4$ and $\lambda_1 + \mu_8 + 6 = 0$. Find a. ***(5 marks)***

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More Questions in Past Papers

1. ***[A.Math 1994 Paper II Q.3]***

(a) Expand $(1 - 2x)^3$ and $(1 + \frac{1}{x})^5$.

(b) Find, in the expansion of $(1 - 2x)^3 (1 + \frac{1}{x})^5$,

(i) the constant term,

(ii) the coefficient of x. ***(5 marks)***

2. ***[A.Math 1981 Paper I Q.1]***

Find the coefficient of x^2 in the expansion of $(1 + 2x)^4 (1 - x)^7$.

(5 marks)

3. ***[A.Math 1989 Paper II Q.1]***

Find the constant term in the expansion of $(1+x)^{10}(1-\frac{2}{x})^3$. ***(5 marks)***

4. ***[A.Math 1988 Paper II Q.1]***

Given $(1 + 3x)^4 (1 - 2x)^5 = 1 + ax + bx^2 +$ higher powers of x, find the values of the constants a and b.

(5 marks)

5. ***[A.Math 1983 Paper I Q.4]***

Expand $(1 + ax)^4 (1 - 4x)^3$ in ascending powers of x up to the term x^2.

Given that the coefficient of x is zero, find the coefficient of x^2.

(7 marks)

6. ***[A.Math 1997 Paper II Q.8]***

Expand $(1 + x)^n (1 - 2x)^4$ in ascending powers of x up to the term x^2, where n is a positive integer.

If the coefficient of x^2 is 54, find the coefficient of x.

(7 marks)

7. ***[A.Math 1984 Paper II Q.1 Modified]***

In the expansion of $(x^2 + \frac{a}{x})^8$, where $a \neq 0$, the coefficient of x^r is denoted by B_r.

Find the value of a if $B_7 = 4B_{10}$.

(5 marks)

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